

SAC vs. SAC+JEPA at a 10,000-transition budget

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Both runs share the standardized centered-Gaussian terrain, periodic tiling, random start positions, the $P_{\max} = 6$ longitudinal power cap, 16 parallel environments, the same replay configuration, the same evaluation interval, and the held-out terrain seeds 100001–100004. Only the actor objective differs. All numbers below are single-run, 10k-transition, RTX 3090 results.

1 What differs

Aspect	SAC (vanilla)	SAC+JEPA
Actor objective	$\mathbb{E}[\alpha \log \pi(a o) - \min(Q_1, Q_2)]$	$\mathbb{E}[\alpha \log \pi(a o) - \min(Q_1, Q_2)] + \lambda_{\text{JEPA}} \mathcal{L}_{\text{JEPA}} + c_{\text{var}} \mathcal{L}_{\text{var}}$
Predictive term	none	one-step latent MSE: $\ q_{\psi}([f_{\theta}(o_t), a_t]) - \text{sg}[\bar{f}(o_{t+1})]\ _2^2$
λ_{JEPA}	0	0.1 (measured run); 0.025 (corrected overlay)
EMA target cadence	not applicable	once per replay-gradient update (measured run, unguarded); once per 64 collected transitions (corrected)
Variance floor	none	none (measured run); $0.01 \text{relu}(0.1 - \text{std}(f_{\theta}))$ (corrected)
Reported metrics	<code>actor_loss</code> , <code>critic_loss</code> , <code>alpha</code>	adds <code>actor_total_loss</code> , <code>jepa_loss</code> , and (corrected) <code>jepa_variance_loss</code> , <code>jepa_embedding_std</code>
Replay, collection, terrain, seeds	shared	shared

2 Result

Transitions	SAC	SAC+JEPA
2,048	0.064 ± 0.031	0.064 ± 0.031
4,096	0.190 ± 0.027	0.190 ± 0.029
6,144	0.292 ± 0.042	0.279 ± 0.041
8,192	0.673 ± 0.041	0.641 ± 0.048
10,000	1.655 ± 0.056	1.278 ± 0.042

The endpoints differ by 0.377 (22.8%) in favour of vanilla SAC. The $\pm$ values are standard deviations across the four held-out terrain seeds, *not* across training runs, so they are not an error bar on the SAC-vs-SAC+JEPA difference; with one training seed per arm this gap is not statistically distinguishable.

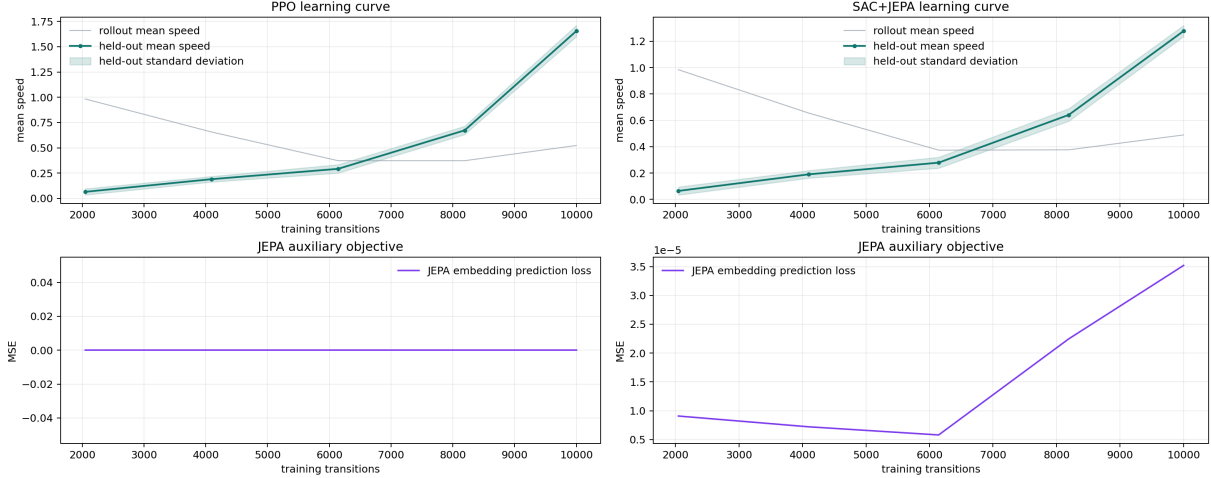


Figure 1: Left: SAC-only. Right: SAC+JEPA ($\lambda_{\text{JEPA}} = 0.1$, pre-correction branch). Both are at the same 10k budget.

3 Why SAC+JEPA is below vanilla

1. **The measured run used the pre-correction EMA cadence.** The target was updated once per replay-gradient update. With 16 environments and one update per transition, roughly 1,056 EMA calls occur between transitions 1,000 and 2,048, so $1 - (1 - 0.01)^{1056} \approx 99.998\%$ of the online weights are absorbed: the target is numerically the online encoder. The objective then reduces to a near-identity map, which is exactly what the observed loss of 9.1×10^{-6} to 3.5×10^{-5} indicates.
2. **One-step horizon.** With $dt = 0.02$ and two substeps, a step advances 0.04s, so consecutive field-of-view observations are nearly identical. Even with a correctly lagged target, “predict z_{t+1} from z_t ” is close to trivial and carries little dynamics signal.
3. **The coefficient was inherited from PPO.** PPO+JEPA reached a loss of 5.95×10^{-4} at the same λ , about $17\times$ the SAC loss scale, so $\lambda = 0.1$ is not matched to the SAC encoder.
4. **Placement.** JEPA shapes only the actor encoder. The twin Q critics keep independent encoders and receive no auxiliary signal, so the term cannot improve value estimation and can only perturb the policy representation.
5. **Budget and variance.** At 10k transitions both curves are inside their steep takeoff (SAC rises $0.67 \rightarrow 1.65$ between 8,192 and 10,000), so a one-step shift in takeoff timing moves the endpoint by roughly the observed margin.

4 What the corrected branch changes, and what remains

The corrected code (i) updates the EMA target once per 64 collected transitions, decoupling the lag from `num_parallel_envs` and `updates_per_step` (effective lag ≈ 637 gradient steps instead of ≈ 100); (ii) adds a VICReg-style variance floor on the online representation; (iii) logs the unregularized actor objective separately from the optimized combined objective; and (iv) begins at $\lambda_{\text{JEPA}} = 0.025$.

Two limitations remain. First, **the JEPA term is negligible relative to the variance term** at the corrected scales: at the observed 10k loss of 9.1×10^{-6} the JEPA contribution is 2.3×10^{-7} while the variance term is $\approx 5 \times 10^{-4}$, a ratio of roughly 2,200 (and still ≈ 570 at the largest observed loss). A corrected run therefore tests a variance regularizer more than it tests JEPA. Second, the trainer *raises* when $\lambda_{\text{JEPA}} > 0$ and $c_{\text{var}} \leq 0$, so the variance term cannot be ablated, and the same guard is absent from the PPO trainer. The horizon is still one step and the objective is still attached only to the actor encoder.

Status. No 10k run of the corrected branch exists yet. The 1.278 result above came from the earlier branch, so it is evidence that the original auxiliary path ran, not a test of the corrected objective. Multi-seed runs at ≥ 100 k transitions, with the variance term ablated, are required before any performance conclusion.